Supplementary Information for "The Tritone, the Minor Sixth and the Dissonance Curve: A Reanalysis of Dense Two-Note Ratings"

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Section, table and figure numbers prefixed with S refer to this document; others refer to the main text. Analyses chosen after results had been seen are labelled exploratory where they are introduced (main-text Table 3 says when each analysis was fixed); all analyses in Sections S8–S13 are exploratory unless stated otherwise. Units are within-listener SD unless stated; 95% CIs are in square brackets. Every table and figure in this document shows my analysis of published data (this study); no new listener data were collected.

S1 Model implementation

This section gives the formulas and constants that the main text describes in words (main-text Section 5). All models are published models; the implementations are mine and were checked against the published ones as described at the end of the section.

Notation. A sound is a set of partials with frequencies f_i (in Hz) and amplitudes a_i. For a dyad the partials of both tones are pooled into one set.

Hutchinson–Knopoff roughness (H&K). For two partials with mean frequency $\bar{f}$ and separation Δf, the separation is expressed in critical bandwidths, $y = \Delta f / \mathrm{CBW}(\bar{f})$ with $\mathrm{CBW}(\bar{f}) = 1.72\,\bar{f}^{\,0.65}$. The roughness of the pair is $g(y) = \left((y/0.25)\,e^{1-y/0.25}\right)^2$ for $y \le 1.2$ and zero beyond; g is zero at $y = 0$ and reaches its maximum of 1 at $y = 0.25$. The roughness of the sound is $\sum_{i<j} a_i a_j\, g(y_{ij}) / \sum_i a_i^2$ (Hutchinson & Knopoff, 1978).

Sethares. The Sethares model replaces g by a difference of two exponentials in the frequency separation, scaled by the lower frequency of the pair, with the constants published by Sethares (1993); the code is in `core/dissonance_model.py`.

Revised H&K and the composite of Marjieh et al. Marjieh et al. (2024) change H&K in two ways. Below p critical bandwidths the pair's value blends g with a negative term of strength q, so that very slow beating counts as pleasant; and the amplitude weight becomes $(a_i a_j)^{r/2}$, normalised by $\sum_i a_i^r$. Their fitted constants are $p = 0.096$, $q = 1.632$ and $r = 1.359$. Their composite is harmonicity times 0.837 minus revised roughness. I transcribed the model from the authors' code (Harrison, 2023) and did not refit any constant.

Harmonicity. The model of Harrison and Pearce (2018, 2020) works on *pitch classes*: a frequency is reduced to its position within the octave, measured in cents from 0 to 1200. (i) Each partial adds a Gaussian bump, centred on its pitch class and with an SD equal to the tolerance, to a spectrum of 1200 one-cent bins; the published tolerance is 6.83 cents and the wider one used in this paper is 20 cents. (ii) This spectrum is compared with a *template*, the pitch-class spectrum of the first 11 harmonics of a single note, by *cross-correlation*: the template is shifted to each of the 1200 possible positions and the overlap with the spectrum is recorded. The result, normalised to sum to one, is the virtual-pitch profile: how strongly each candidate pitch is supported. (iii) Harmonicity is the *Kullback–Leibler (KL) divergence* of that profile from a flat one, a standard measure of how far a distribution is from uniform. A profile with one sharp peak gives a high value. Harmonic i of the template has weight $1/i$ for the dyads, as in Marjieh et al. (2024), with the partial amplitudes of the stimuli; for the chords it has weight $i^{-0.75}$, with idealised tones of 11 harmonics, as in `incon` (Harrison, 2024). I did not tune these weights. The only quantities fitted in this paper are the intercept and the weights that place roughness and harmonicity on the rating scale (main-text Section 6), and the tolerance, chosen from a short list written down before the screen was run (main-text Section 7.4).

Checks of the implementations. With the lower tone at C4, I computed each model's curve for the stimuli of each of the 13 dense dyad experiments and correlated it with the curve published for that experiment by Marjieh et al. (2024). The smallest Pearson correlation over the experiments was $r = 0.9999$ for H&K, 0.9996 for harmonicity and 0.9999 for revised H&K. A value this close to 1 means that the two curves have the same shape to within rounding; it is a check of the code and not a measure of fit to listeners. On the chords of Bowling et al. (2018) my implementations reproduce the values of `incon` exactly when given the same equal-tempered pitches.

S2 Sensitivity of the primary estimand to the excluded window (exploratory)

The fitted curve of the main text leaves out the intervals within ±0.6 semitones of the tritone and of the minor sixth (main-text Section 6). That width is an analyst's choice. As an exploratory check I repeated the held-out analysis of the six US experiments with harmonic spectra with windows of ±0.4 and ±0.8 semitones, with the same bootstrap draws (Table S1). The row for ±0.6 and roughness plus harmonicity is the primary estimand of the main text and reproduces it. The pooled unexplained gap is +0.18 [+0.04, +0.32] with ±0.4, +0.18 [+0.04, +0.32] with ±0.6 and +0.18 [+0.04, +0.32] with ±0.8, and the residuals at the two intervals change only in the second decimal place. The conclusions of the main text therefore do not depend on this choice within the range tried. A window of ±0.8 around the minor sixth extends to 8.8 semitones, close to the major sixth, so wider windows would begin to remove neighbouring intervals from the fit.

S3 Participants per experiment

Table S2 gives three counts for every experiment of Marjieh et al. (2024) used here: the number of participants reported in their article, the number of participant identifiers in the released rating file, and the number analysed in this study. The released files are complete in ratings, but for the US experiments they contain fewer identifiers than the article reports participants, and for the Korean experiments more (the article reports 68 Korean participants in all). The

Table S1: The held-out residual at the tritone and at the minor sixth, and the unexplained gap, for three widths of the window excluded from the fit (exploratory). Random-effects pooled estimates over the six US experiments with harmonic spectra [Hartung–Knapp 95% interval], SD units.

Window	Curve	Residual, tritone	Residual, minor sixth	Unexplained gap
±0.4	Roughness	−0.09 [−0.21, +0.03]	+0.17 [+0.06, +0.29]	+0.26 [+0.14, +0.37]
	Roughness + harmonicity (20 cents)	−0.02 [−0.14, +0.11]	+0.17 [+0.03, +0.31]	+0.18 [+0.04, +0.32]
±0.6	Roughness	−0.09 [−0.22, +0.03]	+0.17 [+0.06, +0.29]	+0.26 [+0.14, +0.37]
	Roughness + harmonicity (20 cents)	−0.02 [−0.15, +0.11]	+0.17 [+0.03, +0.31]	+0.18 [+0.04, +0.32]
±0.8	Roughness	−0.09 [−0.22, +0.04]	+0.18 [+0.06, +0.29]	+0.26 [+0.14, +0.37]
	Roughness + harmonicity (20 cents)	−0.01 [−0.14, +0.12]	+0.17 [+0.03, +0.31]	+0.18 [+0.04, +0.32]

article's Methods section states that each Korean participant was allowed to take part in the same experiment up to three times, which would account for the larger number if an identifier is one sitting and not one person. For the smaller numbers in the US files I found no explanation in the article, its Supplementary Information or the repository. This study treats each identifier as one participant. The only exclusion made in this study is of identifiers with the same rating on every trial, for which the standardisation of the ratings is undefined.

Table S2: Participants per experiment: reported by Marjieh et al. (2024) (prior data; US experiments from the text of the article, Korean experiments from its Supplementary Table 1, which gives 202 and not 198 for the piano experiment), identifiers present in the released rating file, identifiers dropped here because every rating was the same, and identifiers analysed (this study).

Experiment	Reported	Released file	Dropped	Analysed
United States				
Harmonic, 3 dB/oct	198	165	5	160
5 equal harmonics	149	147	0	147
5 harmonics, no 3rd	160	151	0	151
Flute	190	188	0	188
Guitar	210	191	5	186
Piano	198	194	3	191
Pure tones	176	156	5	151
Bonang (harmonic bass)	170	152	6	146
Stretched (octave 2.1)	194	157	4	153
Compressed (octave 1.9)	202	159	5	154
South Korea				
Harmonic, 3 dB/oct	24	40	0	40
Stretched (octave 2.1)	20	37	0	37
Compressed (octave 1.9)	24	45	0	45
Roll-off experiment (United States)				
Roll-off varied from trial to trial	322	301	6	295

S4 What adding harmonicity changes at the tritone and the minor sixth (exploratory)

The main text reports the held-out residual at the tritone and at the minor sixth under roughness alone and under roughness plus harmonicity with the 20-cent tolerance (main-text Table 5). Table S3 gives the difference between the two curves directly: the residual under roughness plus harmonicity minus the residual under roughness alone, in each experiment, with the same held-out fits and the same bootstrap draws as the main-text table. Because the observed ratings are the same under both curves, this difference is the change in the curve's prediction, with its

sign reversed, and its interval is narrow. The analysis was added after the results of the main text had been seen. With the published tolerance the pooled changes are +0.05 [+0.03, +0.06] at the tritone, −0.01 [−0.05, +0.02] at the minor sixth and −0.06 [−0.09, −0.03] in the gap.

Table S3: Change in the held-out residual when harmonicity (20-cent tolerance) is added to roughness: at the tritone, at the minor sixth, and in the unexplained gap (exploratory). SD units [participant-bootstrap 95% CI]; pooled row: random effects over the six US experiments with harmonic spectra [Hartung–Knapp 95% interval].

Experiment	Tritone, $r(6)$	Minor sixth, $r(8)$	Gap, $r(8) - r(6)$
Harmonic, 3 dB/oct	+0.07 [+0.06, +0.08]	+0.01 [+0.01, +0.01]	−0.06 [−0.07, −0.05]
5 equal harmonics	+0.08 [+0.07, +0.09]	−0.03 [−0.04, −0.03]	−0.11 [−0.13, −0.10]
5 harmonics, no 3rd	+0.06 [+0.05, +0.07]	−0.07 [−0.08, −0.05]	−0.12 [−0.14, −0.10]
Flute	+0.09 [+0.07, +0.10]	+0.02 [+0.01, +0.02]	−0.07 [−0.08, −0.06]
Guitar	+0.07 [+0.06, +0.08]	+0.03 [+0.03, +0.04]	−0.04 [−0.05, −0.03]
Piano	+0.08 [+0.07, +0.09]	+0.03 [+0.02, +0.03]	−0.06 [−0.06, −0.05]
Pure tones	+0.05 [+0.04, +0.06]	+0.03 [+0.03, +0.04]	−0.02 [−0.02, −0.02]
Pooled, harmonic (RE, $k = 6$)	+0.07 [+0.06, +0.08]	0.00 [−0.04, +0.04]	−0.07 [−0.11, −0.04]

S5 Pooling: DerSimonian–Laird and Hartung–Knapp intervals

The main text pools experiments by random effects with the DerSimonian–Laird estimate of τ^2 and a Hartung–Knapp interval. Table S4 gives both intervals for the rows of Table S23. Of 326 random-effects estimates in the released results, 22 change whether their interval excludes zero when the Hartung–Knapp interval replaces the DerSimonian–Laird one; the list is in `results/re_intervals.json`. Table S5 gives the dip, observed and left unexplained, for every experiment, with each model fitted to the experiment itself.

Table S4: Random-effects pooled estimates with the DerSimonian–Laird (DL, normal) and Hartung–Knapp (t distribution with $k - 1$ degrees of freedom) 95% intervals, SD units, each model fitted to the experiment itself. Harmonic: six US experiments with harmonic spectra; eight: adds pure tones and the bonang.

	Gap, harmonic			Gap, eight			Dip, harmonic		
	Est.	DL	Hartung–Knapp	Est.	DL	Hartung–Knapp	Est.	DL	Hartung–Knapp
Observed	+0.26	[+0.18, +0.34]	[+0.16, +0.36]	+0.27	[+0.20, +0.33]	[+0.18, +0.35]	+0.30	[+0.24, +0.36]	[+0.22, +0.37]
After roughness (H&K)	+0.26	[+0.17, +0.35]	[+0.15, +0.37]	+0.26	[+0.19, +0.32]	[+0.18, +0.33]	+0.25	[+0.16, +0.33]	[+0.13, +0.36]
After roughness (Sethares)	+0.25	[+0.16, +0.35]	[+0.13, +0.38]	+0.25	[+0.17, +0.33]	[+0.16, +0.34]	+0.28	[+0.20, +0.36]	[+0.17, +0.39]
After harmonicity alone	+0.11	[−0.01, +0.24]	[−0.05, +0.27]	+0.14	[+0.03, +0.25]	[+0.02, +0.27]	+0.03	[−0.07, +0.13]	[−0.10, +0.16]
After roughness + harmonicity	+0.19	[+0.06, +0.32]	[+0.02, +0.36]	+0.20	[+0.10, +0.31]	[+0.08, +0.32]	+0.14	[+0.02, +0.25]	[−0.02, +0.29]
harmonicity template $i^{-0.75}$ (`incon`)	+0.20	[+0.08, +0.32]	[+0.05, +0.36]	+0.21	[+0.12, +0.31]	[+0.10, +0.32]	+0.15	[+0.04, +0.26]	[0.00, +0.30]
After revised H&K + harmonicity	+0.28	[+0.15, +0.40]	[+0.11, +0.44]	+0.27	[+0.17, +0.36]	[+0.15, +0.38]	+0.20	[+0.07, +0.34]	[+0.02, +0.39]
After Marjieh et al. composite[a]	+0.17	[+0.06, +0.28]	[+0.03, +0.31]	+0.18	[+0.09, +0.27]	[+0.08, +0.28]	+0.05	[−0.08, +0.19]	[−0.13, +0.23]

S6 The screen: rows examined afterwards

Table S6 lists the models examined after the screen of main-text Section 7.4, in the format of main-text Table 6 (exploratory). The first row extends the tolerance grid beyond the values listed before the screen; the held-out R^2 changes little over a broad range (main-text Figure 3), so the choice of 20 cents is representative rather than an optimum. The other rows use the timbre-relative harmonicity of Section S9, alone or added to the 20-cent term.

Table S5: Mean of the fourth and fifth minus the tritone (the dip), observed and left unexplained [95% participant-bootstrap CI], SD units, each model fitted to the experiment itself. Bold: CI excludes zero. The composite of Marjieh et al. was fitted to these data.

			Left unexplained after		
Experiment	N	Observed	Roughness	Roughness + harmonicity	Composite
United States					
Harmonic, 3 dB/oct	160	**+0.38 [+0.25, +0.51]**	**+0.29 [+0.17, +0.42]**	+0.10 [−0.02, +0.22]	**+0.15 [+0.03, +0.27]**
5 equal harmonics	147	**+0.27 [+0.14, +0.40]**	+0.12 [−0.02, +0.24]	−0.09 [−0.21, +0.04]	**−0.13 [−0.26, −0.01]**
5 harmonics, no 3rd	151	**+0.23 [+0.14, +0.33]**	**+0.28 [+0.19, +0.38]**	**+0.23 [+0.14, +0.32]**	**+0.21 [+0.12, +0.31]**
Flute	188	**+0.23 [+0.11, +0.34]**	+0.10 [−0.02, +0.22]	+0.03 [−0.09, +0.15]	**−0.19 [−0.31, −0.08]**
Guitar	186	**+0.33 [+0.20, +0.46]**	**+0.32 [+0.19, +0.45]**	**+0.25 [+0.12, +0.39]**	**+0.14 [+0.01, +0.27]**
Piano	191	**+0.39 [+0.27, +0.51]**	**+0.36 [+0.24, +0.48]**	**+0.28 [+0.17, +0.41]**	**+0.13 [+0.01, +0.25]**
Pure tones	151	**+0.28 [+0.17, +0.39]**	**+0.28 [+0.17, +0.39]**	**+0.20 [+0.10, +0.32]**	+0.10 [−0.01, +0.22]
Bonang (harmonic bass)	146	**+0.36 [+0.23, +0.49]**	**+0.30 [+0.18, +0.42]**	+0.04 [−0.09, +0.17]	**+0.13 [+0.01, +0.25]**
Stretched (octave 2.1)	153	−0.03 [−0.15, +0.09]	−0.03 [−0.15, +0.08]	−0.03 [−0.15, +0.08]	−0.04 [−0.15, +0.08]
Compressed (octave 1.9)	154	+0.03 [−0.09, +0.14]	+0.03 [−0.09, +0.14]	+0.03 [−0.08, +0.15]	+0.02 [−0.09, +0.13]
South Korea					
Harmonic, 3 dB/oct	40	**+0.25 [+0.10, +0.42]**	**+0.19 [+0.05, +0.35]**	−0.04 [−0.17, +0.09]	+0.07 [−0.08, +0.21]
Stretched (octave 2.1)	37	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]
Compressed (octave 1.9)	45	−0.01 [−0.15, +0.15]	0.00 [−0.15, +0.16]	+0.04 [−0.11, +0.20]	0.00 [−0.14, +0.16]

Table S6: Models examined after the screen (exploratory; leave one US experiment out; point estimates). Columns as in main-text Table 6.

	Held-out R^2					Stretched, dip left	
Model	10 US	6 harm.	Gap left	Dip left	TT − other	at 6.00	at 6.42
H&K + harmonicity, σ grid extended to 60 cents	0.542	0.657	+0.19	+0.14	+0.16	−0.03	+0.09
H&K + timbre-relative harmonicity	0.515	0.640	+0.20	+0.19	+0.18	−0.06	+0.02
H&K + harmonicity(20) + timbre-relative(20)	0.534	0.644	+0.21	+0.14	+0.16	−0.02	+0.10

S7 The dip at every interval

Figure S1 repeats the dip analysis at every integer interval, pooled over the eight US experiments without stretched or compressed partials (fixed-effect, inverse-variance weights; harmonicity at the published 6.83-cent tolerance). It is given for comparison with the random-effects residual profile of Table S25. Observed dips are largest at the minor second (+0.50) and the minor ninth (+0.41); the tritone ranks 4 of 14 (+0.30). After fitted roughness the tritone has the largest unexplained dip (+0.26; next, the minor ninth, +0.19). After roughness and harmonicity it ranks 2 (+0.15, behind the major second at +0.19). A dip compares an interval with its two neighbours, so a large dip at the tritone after harmonicity can come from the fourth and fifth rated above the curve rather than from the tritone rated below it; the residual profile of the main text shows that it does.

S8 The in-tune bonus and the step

All analyses in this section are exploratory; the patterns were noticed in the residual profile. Three curves are compared: M1, H&K roughness plus harmonicity at 20 cents; M2, M1 plus a linear term in interval size (the curve used in the main text); M3, M1 plus the sharpness proxy of Section S12. For each, the held-out residual $r(c)$ is split at each equal-tempered interval c into a local bonus $b(c) = r(c) - [r(c-0.5) + r(c+0.5)]/2$ and a baseline $q(c) = [r(c-0.5) + r(c+0.5)]/2$, so that $r(c) = b(c) + q(c)$ and the gap $r(8) - r(6)$ splits into a bonus part $b(8) - b(6)$ and a baseline part $q(8) - q(6)$. The step is the coefficient of an indicator for $c \geq 8$ in a regression of $q(c)$ on c over 1–11 semitones (or 1–14), or the increment $q(8) - q(7)$ minus the mean of the other increments.

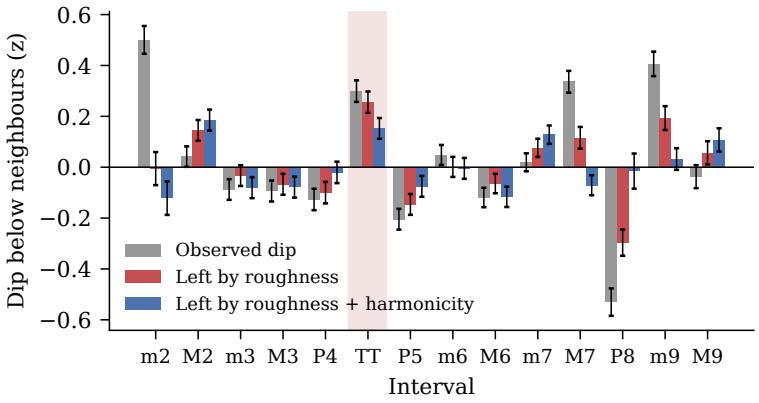


Figure S1: Dip of each integer interval below the mean of its two semitone neighbours, pooled over eight US experiments with fixed-effect (inverse-variance) weights; bars, ±1.96 pooled SE. Grey: observed. Red: left by fitted H&K roughness. Blue: left by roughness plus harmonicity at the published 6.83-cent tolerance. Each interval's fit excludes its own window. Shaded: tritone.

Plain means and random effects. Table S7 gives every quantity under both a plain mean over the six harmonic spectra, with a participant-bootstrap CI that treats the spectra as fixed, and random-effects pooling with a Hartung–Knapp CI, which treats them as a sample. The two disagree. Under plain means the bonus part (M2: +0.07 [+0.01, +0.13]), the baseline part (+0.15 [+0.11, +0.19]), the step (+0.10 [+0.06, +0.14]) and the tritone's bonus shortfall relative to the other intervals (+0.07 [+0.03, +0.11]) all exclude zero, as they do under M1 and M3. Under random effects the bonus part (+0.06 [−0.03, +0.16]), the step as an indicator (+0.10 [−0.03, +0.23]) and the tritone's shortfall (+0.07 [−0.01, +0.14]; M1 +0.07 [0.00, +0.14], M3 +0.07 [0.00, +0.14]) include zero, while the gap (+0.21 [+0.06, +0.36]), the baseline part (+0.15 [+0.02, +0.28]) and the step as an increment (+0.10 [+0.01, +0.19]) do not. The main text reports the random-effects version, as for the primary estimand, because the question is about spectra in general. The plain-mean figures answer a narrower question, about these six spectra, and they alone would suggest that the tritone receives about half the bonus; the random-effects version does not support that statement. The same plain-mean contrast computed on the curve with the sharpness term is +0.07 [+0.02, +0.11].

One contrast holds under both: the major second and minor seventh receive a smaller bonus than the other eight intervals from the minor second to the major seventh (random effects −0.12 [−0.14, −0.10]; plain −0.12 [−0.15, −0.09]), and the tritone does not differ from them (+0.03 [−0.04, +0.09]). This contrast was chosen after seeing the profile. The individual bonuses under M2 (plain means, point estimates) are +0.14, +0.05, +0.18, +0.20, +0.16, +0.07, +0.19, +0.14, +0.15, +0.04 and +0.13 from the minor second to the major seventh.

The step across analysis choices and spectra. Table S8 repeats the step under M2 with other kernel widths, exclusion windows and raw 0.25-semitone bins. The plain-mean step is stable across these choices (kernel SD 0.1: +0.12 [+0.07, +0.17]; 0.3: +0.08 [+0.05, +0.11]; window ±0.4: +0.09 [+0.06, +0.13]; ±0.8: +0.10 [+0.07, +0.13]; bins: +0.13 [+0.07, +0.18]), but in every variant it comes mainly from the guitar and piano, whose spectra are proxies from one analysed tone each, and it is absent for the synthetic harmonic tone and the flute.

Table S7: Decomposition of the held-out gap under three curves (exploratory). For each: plain mean over the six harmonic spectra [participant-bootstrap CI] and random effects [Hartung–Knapp CI]. M1: H&K + harmonicity (20 cents); M2: + interval size; M3: + sharpness proxy. Bold: CI excludes zero.

	M1		M2		M3	
Quantity	Plain	Random effects	Plain	Random effects	Plain	Random effects
Gap left, $r(8) - r(6)$	**+0.19 [+0.13, +0.25]**	**+0.18 [+0.04, +0.32]**	**+0.22 [+0.16, +0.28]**	**+0.21 [+0.06, +0.36]**	**+0.21 [+0.16, +0.27]**	**+0.21 [+0.06, +0.35]**
Bonus part, $b(8) - b(6)$	**+0.07 [+0.01, +0.12]**	+0.06 [−0.03, +0.16]	**+0.07 [+0.01, +0.13]**	+0.06 [−0.03, +0.16]	**+0.07 [+0.01, +0.13]**	+0.06 [−0.03, +0.16]
Baseline part, $q(8) - q(6)$	**+0.12 [+0.09, +0.16]**	+0.12 [−0.01, +0.25]	**+0.15 [+0.11, +0.19]**	**+0.15 [+0.02, +0.28]**	**+0.15 [+0.11, +0.19]**	**+0.14 [+0.01, +0.27]**
Step (indicator, 1–11)	**+0.06 [+0.02, +0.10]**	+0.06 [−0.06, +0.18]	**+0.10 [+0.06, +0.14]**	+0.10 [−0.03, +0.23]	**+0.10 [+0.06, +0.13]**	+0.10 [−0.03, +0.22]
Step (indicator, 1–14)	**+0.11 [+0.07, +0.14]**	**+0.10 [+0.01, +0.20]**	**+0.13 [+0.10, +0.16]**	**+0.13 [+0.03, +0.23]**	**+0.12 [+0.09, +0.16]**	**+0.12 [+0.02, +0.22]**
Step (7→8 increment)	**+0.10 [+0.07, +0.12]**	**+0.10 [+0.01, +0.18]**	**+0.10 [+0.07, +0.13]**	**+0.10 [+0.01, +0.19]**	**+0.10 [+0.07, +0.13]**	**+0.10 [+0.01, +0.19]**
Mean bonus, other intervals	**+0.14 [+0.12, +0.15]**	**+0.13 [+0.08, +0.19]**	**+0.14 [+0.12, +0.15]**	**+0.14 [+0.08, +0.19]**	**+0.14 [+0.12, +0.15]**	**+0.14 [+0.08, +0.19]**
Tritone bonus	**+0.07 [+0.03, +0.11]**	+0.06 [0.00, +0.13]	**+0.07 [+0.03, +0.11]**	+0.07 [0.00, +0.14]	**+0.07 [+0.03, +0.11]**	+0.07 [0.00, +0.14]
Other − tritone bonus	**+0.07 [+0.03, +0.11]**	**+0.07 [0.00, +0.14]**	**+0.07 [+0.03, +0.11]**	+0.07 [−0.01, +0.14]	**+0.07 [+0.03, +0.11]**	+0.07 [0.00, +0.14]
Tritone − M2/m7 bonus	+0.02 [−0.03, +0.07]	+0.02 [−0.05, +0.08]	+0.03 [−0.02, +0.08]	+0.03 [−0.04, +0.09]	+0.03 [−0.02, +0.07]	+0.02 [−0.04, +0.09]
M2/m7 − other bonus	**−0.11 [−0.14, −0.08]**	**−0.11 [−0.13, −0.09]**	**−0.12 [−0.15, −0.09]**	**−0.12 [−0.14, −0.10]**	**−0.12 [−0.15, −0.09]**	**−0.12 [−0.14, −0.10]**

Table S8: The step (M2) under other analysis choices (exploratory): plain mean over the six harmonic spectra [participant-bootstrap CI] of the step as an indicator and as an increment, the bonus and baseline parts of the gap, and the step per spectrum (point estimates). Guitar and piano use proxy spectra. Bold: CI excludes zero.

	Six harmonic spectra				Step per spectrum					
Variant	Step	Increment	Bonus part	Baseline part	Harm.	5 eq.	No 3rd	Flute	Guitar	Piano
Default (SD 0.2, ±0.6)	**+0.10 [+0.07, +0.13]**	**+0.10 [+0.07, +0.13]**	**+0.07 [+0.01, +0.12]**	**+0.15 [+0.12, +0.18]**	−0.02	+0.04	+0.09	+0.01	+0.31	+0.17
Kernel SD 0.1	**+0.12 [+0.07, +0.17]**	**+0.10 [+0.06, +0.14]**	**+0.09 [0.00, +0.19]**	**+0.14 [+0.10, +0.19]**	+0.03	+0.03	+0.11	−0.01	+0.35	+0.22
Kernel SD 0.3	**+0.08 [+0.05, +0.11]**	**+0.09 [+0.07, +0.12]**	**+0.04 [+0.01, +0.07]**	**+0.15 [+0.12, +0.18]**	−0.03	+0.02	+0.07	+0.02	+0.27	+0.14
Exclusion ±0.4	**+0.09 [+0.06, +0.13]**	**+0.10 [+0.07, +0.13]**	**+0.07 [+0.01, +0.12]**	**+0.15 [+0.11, +0.18]**	−0.03	+0.04	+0.08	0.00	+0.31	+0.16
Exclusion ±0.8	**+0.10 [+0.07, +0.13]**	**+0.10 [+0.07, +0.13]**	**+0.07 [+0.01, +0.12]**	**+0.15 [+0.12, +0.18]**	−0.02	+0.04	+0.09	+0.01	+0.31	+0.17
Raw 0.25-semitone bins	**+0.13 [+0.07, +0.18]**	**+0.11 [+0.07, +0.16]**	+0.08 [−0.03, +0.18]	**+0.15 [+0.09, +0.20]**	+0.03	+0.03	+0.12	0.00	+0.34	+0.25

S9 Stretched and compressed partials

The dip at the stretched scale. Repeating the analysis at the stretched scale's own fourth, tritone, fifth and minor sixth, the stretched tones show an observed dip of +0.19 [+0.07, +0.31] at 6.42 semitones (paired difference from the dip at 6, on the same bootstrap draws, +0.22 [+0.07, +0.37]). Fitted H&K roughness fitted to the stretched experiment itself leaves +0.11 [−0.01, +0.23] of it, and adding harmonicity changes little (+0.11 [−0.02, +0.23]). The gap to the stretched minor sixth is +0.06 [−0.07, +0.20]. For compressed tones the dip at 5.56 is +0.07 [−0.05, +0.19] (paired difference +0.04 [−0.14, +0.22]). For harmonic tones the procedure returns the ordinary estimates (dip +0.38 [+0.26, +0.50]).

The bonus at the stretched scale (A4). If the in-tune bonus were attached to intervals as heard relative to the heard octave, it should appear at the stretched scale's own intervals. Table S9 gives the mean bonus of the intervals other than the tritone at the scale matched to each spectrum and at the equal-tempered positions, with the minimum effect detectable with 80% power (2.8 SE). For US listeners the bonus is +0.13 [+0.10, +0.17] for harmonic tones and +0.02 [−0.02, +0.05] and +0.01 [−0.02, +0.05] at the stretched and compressed scales, with detectable effects of 0.05 and 0.05. The absence is therefore not a matter of power.

Timbre-relative harmonicity. The published harmonicity model cannot follow the shifted dip, because its template is a harmonic series. As an exploratory step taken after the screen, I replaced the template with the upper tone's own partials and let the pitch-class circle wrap at the tone's own octave, one way of reading the proposal that harmonic templates are learned from the sounds heard (Shamma & Klein, 2000). Out of sample, this model predicts a dip of +0.17 at 6.42 semitones for the stretched tones and leaves +0.02 [−0.11, +0.14], against +0.08 [−0.05, +0.20] with the standard template (paired difference −0.06 [−0.07, −0.05]); Marjieh et al.'s revised roughness also leaves little there (−0.01). With one stretched experiment neither residual is distinguishable from zero. The model pays for this on harmonic tones: the dip left

Table S9: Mean in-tune bonus of the intervals other than the tritone (exploratory; M2 curve) at the scale matched to each spectrum and at equal-tempered positions [participant-bootstrap CI], and the minimum detectable effect (MDE, 2.8 SE). Bold: CI excludes zero.

Spectrum (cohort)	Own scale	Equal-tempered	MDE
Harmonic (US)	**+0.13 [+0.10, +0.17]**	–	0.05
Stretched 2.1 (US)	+0.02 [−0.02, +0.05]	−0.01 [−0.04, +0.02]	0.05
Compressed 1.9 (US)	+0.01 [−0.02, +0.05]	+0.03 [−0.01, +0.06]	0.05
Harmonic (Korea)	**+0.07 [+0.03, +0.12]**	–	0.06
Stretched 2.1 (Korea)	−0.04 [−0.08, +0.01]	0.00 [−0.04, +0.04]	0.07
Compressed 1.9 (Korea)	−0.01 [−0.05, +0.04]	+0.04 [0.00, +0.09]	0.06

at 600 cents rises to +0.19 (standard template +0.14; paired difference +0.05 [+0.04, +0.06]) and the held-out R^2 falls to 0.515 (paired difference −0.018 [−0.037, −0.001]). Adding it to the 20-cent term does not improve the fit (paired R^2 difference −0.014 [−0.022, −0.008]). In summary, a template matched to the timbre can reproduce the stretched-scale dip but is not a better general curve, and a single stretched spectrum cannot decide between it and revised roughness.

S10 Roll-off

Table S10 gives the roll-off experiment at the three pre-set roll-off levels. The roughness curve predicts almost no gap at any roll-off, and the observed gap does not shrink as the upper partials fade.

Table S10: Study 3 of Marjieh et al. (2024) (harmonic tones, roll-off varied continuously; $N = 295$), smoothed at three roll-off levels (kernel SD 1.5 dB/octave). Pred.: gap predicted by the fitted roughness curve. SD units [95% CI]. Bold: CI excludes zero.

dB/oct	Observed gap	Roughness pred.	After roughness	After rough. + harm.	Observed dip	Dip after rough. + harm.
2	**+0.27 [+0.10, +0.44]**	+0.01 [+0.01, +0.02]	**+0.26 [+0.09, +0.43]**	**+0.20 [+0.04, +0.37]**	**+0.41 [+0.26, +0.57]**	**+0.20 [+0.04, +0.34]**
7	**+0.18 [+0.05, +0.32]**	0.00 [−0.01, +0.01]	**+0.18 [+0.05, +0.32]**	**+0.13 [0.00, +0.26]**	**+0.35 [+0.23, +0.47]**	**+0.14 [+0.02, +0.26]**
12	**+0.29 [+0.16, +0.43]**	0.00 [0.00, 0.00]	**+0.29 [+0.16, +0.43]**	**+0.25 [+0.12, +0.38]**	**+0.36 [+0.21, +0.50]**	**+0.22 [+0.08, +0.36]**

S11 Korean listeners

Korean listeners rating the same harmonic stimuli show a smaller gap than US listeners (difference −0.13 [−0.37, +0.10]; after roughness and harmonicity −0.14 [−0.39, +0.08]). Their dip at the tritone is clear (+0.25 [+0.10, +0.42]) and the two-mechanism curve accounts for it (−0.04). The cohort difference is not specific to harmonic tones: Korean listeners also rate the minor sixth lower relative to the tritone for stretched (−0.15 [−0.36, +0.07]) and compressed (−0.23 [−0.45, −0.02]) partials, where US listeners show no gap.

In-sample fits. Fitted to each Korean experiment alone (in-sample), roughness plus harmonicity at 20 cents reaches R^2 of 0.01–0.12; adding a linear term in interval size raises the mean to 0.87, and adding the sharpness proxy instead raises the range to 0.87–0.93, close to but below the split-half ceilings (0.946–0.950). These in-sample figures overstate what the models predict.

Held-out fits (A3). Table S11 predicts the Korean experiments with weights fitted to all US experiments. The recommended curve does worse than each experiment's mean (mean held-out R^2 −0.19 [−0.26, −0.08]), as does the composite (−0.13 [−0.21, −0.03]); a linear term in

interval size brings the mean to +0.11 [−0.01, +0.22] and the sharpness proxy to +0.24 [+0.09, +0.36]. What the curves miss for Korean listeners is mainly a steeper decline of pleasantness with interval size. Sharpness is one candidate cause; in an online study, differences in playback equipment between cohorts are another.

Table S11: Korean experiments predicted with weights from US listeners (exploratory): held-out R^2 [participant-bootstrap CI], and the split-half ceiling.

Model	Harmonic	Stretched 2.1	Compressed 1.9	Mean
H&K + H (20)	+0.03 [−0.16, +0.23]	−0.15 [−0.28, −0.02]	−0.43 [−0.56, −0.25]	−0.19 [−0.26, −0.08]
+ interval size	+0.28 [+0.09, +0.43]	+0.15 [+0.01, +0.29]	−0.10 [−0.25, +0.08]	+0.11 [−0.01, +0.22]
+ sharpness proxy	+0.40 [+0.21, +0.55]	+0.37 [+0.18, +0.53]	−0.04 [−0.20, +0.14]	+0.24 [+0.09, +0.36]
Composite (Marjieh et al.)	+0.01 [−0.19, +0.19]	−0.11 [−0.20, 0.00]	−0.29 [−0.39, −0.13]	−0.13 [−0.21, −0.03]
Split-half ceiling	0.95	0.95	0.95	

S12 The smooth trend and the sharpness proxy

All analyses in this section are exploratory, except the triad test, whose predictions were written down before the triad data were read.

The proxy. S is the loudness-weighted mean critical-band rate of the partials (weights amplitude$^{0.6}$) with Zwicker's high-frequency weighting, $g(z) = 1$ below 15.8 Bark and $0.066\, e^{0.171z}$ above (Fastl & Zwicker, 2007). It is computed from the synthesis parameters, not from audio, and is not a calibrated sharpness in acum. It was chosen over interval size and the log spectral centroid by the same held-out criterion reported here, so its advantage over them is optimistic.

Pooled weights. Pooling all ten US experiments, the three-term curve with the 20-cent harmonicity tolerance has weights $b_R = -1.84$, $b_H = 0.68$ and $b_S = -0.065$ per unit of S (weighted Bark), in the units of the two-term curve of main-text Section 9.1. I do not propose it as the recommended curve, because the proxy was chosen after seeing the residuals and failed the test whose predictions were written down beforehand (Section S12).

The roll-off check. Fitted separately at five roll-offs (kernel SD 1.5 dB/octave), the slope over interval size of the residual after roughness and harmonicity flattens as the roll-off steepens (Table S12; difference between 1 and 13 dB/octave −0.33 [−0.49, −0.17]). With one weight shared across roll-offs, sharpness fits better than interval size (mean R^2 0.74 against 0.70; difference +0.040 [+0.022, +0.053]; roughness and harmonicity alone 0.49), but the slopes it implies cover only about 50% of the observed flattening. The two terms are strongly correlated, so the sign of the interval-size term with both in the model (0.33 [0.15, 0.54] per octave; sharpness −0.20 [−0.27, −0.15]) is weak evidence. I looked at this experiment before choosing sharpness, so it motivated the hypothesis as much as it tests it.

The triad test. Before reading them, I wrote down (but did not publicly register) three predictions for the one rating set of Marjieh et al. not yet analysed: 80 triads of five equal harmonics with and without the third harmonic, rated by 192 participants. Fitted to the triads, the sharpness weight should be negative; adding sharpness should improve leave-one-chord-out R^2; and the three-term curve with frozen dyad weights should correlate better with the chord

Table S12: Roll-off experiment (exploratory): slope over interval size of the residual after H&K roughness plus harmonicity (20 cents), SD per octave of interval [participant-bootstrap CI], and the slope implied by the sharpness proxy with one shared weight.

Roll-off (dB/octave)	Observed slope	Implied by sharpness
1	−0.45 [−0.60, −0.33]	−0.41
4	−0.41 [−0.51, −0.30]	−0.37
7	−0.39 [−0.48, −0.30]	−0.32
10	−0.29 [−0.40, −0.20]	−0.28
13	−0.13 [−0.22, −0.03]	−0.25

means than the two-term curve. I also recorded the risk that the trend had been absent for five equal harmonics in the dyads. None of the three predictions passed. The fitted weight was negative but its interval included zero (−0.095 [−0.193, +0.010]); leave-one-chord-out R^2 went from 0.748 to 0.753 (+0.005 [−0.009, +0.041]); and the frozen-weight correlation went from 0.874 to 0.880 (+0.006 [−0.006, +0.017]). The triads leave little room for a third term, but they do not confirm it.

S13 Equal temperament and just intonation (A8)

This analysis is exploratory; it was added after the dense-dyad residuals had been seen. It asks whether evaluating the gap at equal-tempered positions creates part of it. With a kernel of SD 0.1 semitones (sensitivity: 0.05 and 0.2), I located, per experiment with a participant-bootstrap CI, the listeners' minor-sixth peak (argmax over 7.5–8.5 semitones), tritone trough (argmin over 5.5–6.5) and, as controls, the peaks of the fifth, major third and major sixth, on the listener profile and on the held-out predictions of each model (Table S13, Figure S2). I then recomputed the gap at equal-tempered positions and at two just-intoned pairs: 8/5 against 7/5 (JIa) and 8/5 against 45/32 (JIb) (Table S14).

Pooled over the six harmonic spectra, the listeners' minor-sixth peak lies at 808 [796, 819] cents, −6 [−17, +6] cents from the peak of the harmonicity model at the published tolerance (813 [812, 815]); at the narrower width it is 805 [797, 813] (difference −8 [−16, 0]). The fifth peaks where the model does (−3 [−6, +1]), and the major third and major sixth lie +8 [0, +17] and +5 [−2, +12] cents from the model's peaks, each displaced from the model's just-intoned peak towards equal temperament, as the minor sixth is, but none of these differences is established. The tritone region is a broad trough (616 [576, 655]) with no distinct minimum at 600 cents or at either just-intoned tritone. The gap left by the recommended curve is +0.20 [+0.02, +0.38] at equal-tempered positions and +0.22 [+0.06, +0.37] and +0.19 [+0.05, +0.33] at the just-intoned pairs; the paired differences, +0.03 [−0.11, +0.17] and +0.01 [−0.10, +0.12], show neither a smaller gap at just intonation nor equivalence within ±0.05 SD. The result is inconclusive. The recommended curve has no interior minor-sixth or major-third peak at this width (rows marked "edge"), so it smooths away a peak that listeners show.

S14 Chords: the three-component decomposition

Table S15 and Figure S3 give the unique variance of interference (H&K), harmonicity and familiarity, each over the other two, in the Bowling and Johnson-Laird chords under four covariate specifications. With chord size controlled and equal-tempered scoring, the unique variance of harmonicity in the Bowling chords (0.037) was about half that of interference (0.075) and familiarity (0.080). Scored at the just intonation actually presented, the three are comparable (0.069,

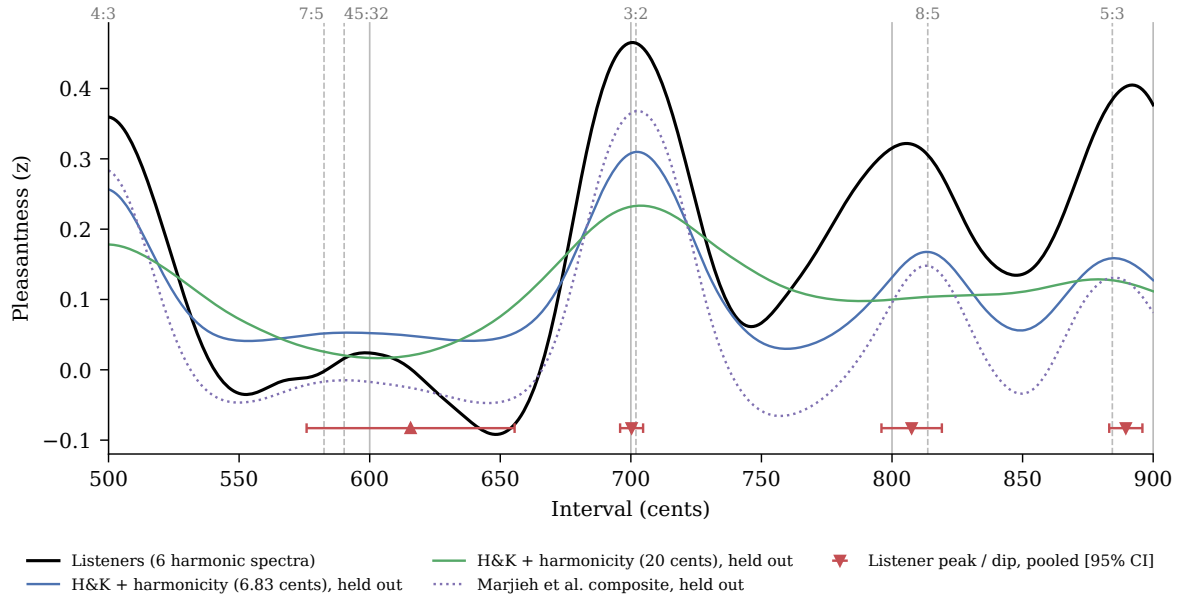


Figure S2: Pooled listener profile of the six harmonic spectra at kernel SD 0.1 semitones (black) with the held-out model curves, from 500 to 900 cents (exploratory). Solid grey lines: equal-tempered positions; dashed: just-intoned ratios. Red markers: pooled listener peak and trough locations [Hartung–Knapp CI].

0.068 and 0.061), and the full model's R^2 rises from 0.738 to 0.775. Without a chord-size control familiarity appears to dominate (0.147 against 0.049 for interference). Across the 298 chords, harmonicity scored in the two tunings correlates $r = 0.941$. The chord bootstraps resample chords, not raters.

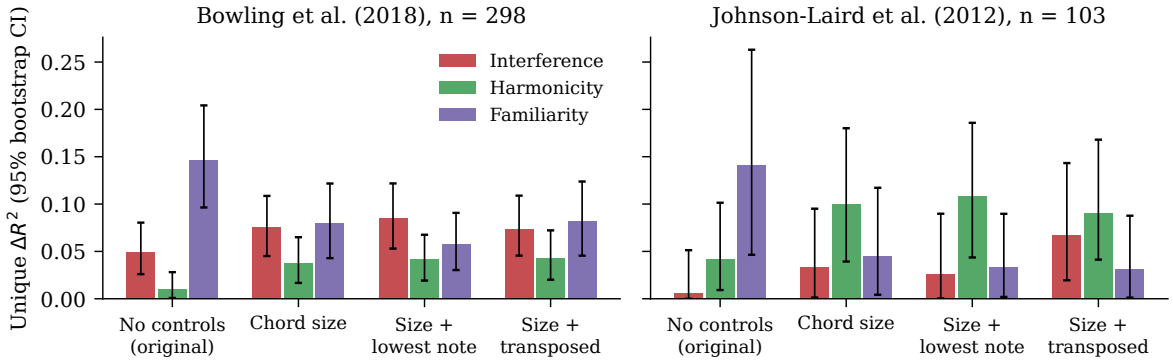


Figure S3: Unique ΔR^2 of interference, harmonicity and familiarity (95% bootstrap CI) under four covariate specifications, models scored in equal temperament.

S15 Roughness curves on measured spectra

Could the problem lie in idealised spectra? A dissonance curve computed from a measured spectrum might place the tritone above the minor sixth. I computed the tritone/minor-sixth roughness difference, $(D_{600} - D_{800})/D_{800}$, for 14 measured spectra under four roughness kernels (Sethares with product and minimum weighting, Vassilakis and H&K). The answer depends on the analysis more than on anything a listener would recognise (Table S16, Figure S4). On one

Philharmonia clarinet spectrum the four kernels give −46.6, −3.4, +6.8 and +39.8%, a range of 86.4 pp. Across the eight C4 instruments the instrument accounts for 55% of the variance of the difference, the kernel for 8% and their interaction for 36% (three stable kernels). The shape constants of the H&K kernel, which are rarely reported, move the difference by a median 13.9 pp over a ±20% grid, and the recording (two clarinet libraries) by a median 6.0 pp (Table S18, Figure S5).

One ordering is robust: under H&K the clarinet rates the minor sixth rougher than the tritone at three registers in both libraries (Iowa: −22.1, −43.0 and −11.7%; Table S17, Figure S6). The reversal comes from a single pair of odd harmonics at the minor sixth (the fifth of the lower note against the third of the upper, 62 Hz apart, 70% of cross-tone roughness) and from the steep decay of the H&K kernel beyond its peak (Figure S7). Depending on instrument and kernel, roughness computed from real spectra can reverse the listeners' ordering as easily as reproduce it.

Extraction. Sustained forte notes were taken from the Philharmonia library (Philharmonia Orchestra, n.d.): cello, clarinet, oboe, flute, bassoon, saxophone and French horn at C4, plus low and high registers for clarinet (D3, C5), cello (C3, C5) and oboe (A♯3, C5); a piano C4 and three further clarinet notes (D3, C4, C5) came from the Iowa library (University of Iowa Electronic Music Studios, n.d.). Each note was analysed with an 8192-point STFT (75% overlap) over the stable middle 60% of the sample. Peaks with prominence above 0.005 of the loudest partial were accepted and the 15 strongest kept; frequencies were refined by parabolic interpolation on log magnitude. The Iowa piano's partials follow the stiff-string law $f_n = nf_0\sqrt{1+Bn^2}$ (Fletcher, 1964) with $B = 3.17 \times 10^{-4}$ (SE 6.3×10^{-7}).

Kernels. Sethares (product): $e^{-3.5s\Delta f} - e^{-5.75s\Delta f}$ with $s = 0.24/(0.0207 f_{\min} + 18.96)$, weighted by $a_1 a_2$ (Sethares, 1993; Schlecht, 2020); Sethares (min): weighted by $\min(a_1, a_2)$ (Sethares, 2005); Vassilakis: weighted by $(a_1 a_2)^{0.1}(2\min(a_1, a_2)/(a_1 + a_2))^{3.11}$ (Vassilakis, 2001); H&K as in main-text Section 5; as a sensitivity check the ERB of Glasberg and Moore (1990) replaced the critical bandwidth. Coincident partials were merged in quadrature. Each spectrum's dyad was swept from unison to the octave in 5-cent steps. Implementations differ between packages (Mashinter, 2006).

Sensitivity factors. Instrument (eight at C4); kernel; H&K constants on a wide grid, $a = 0.15, 0.2, 0.25, 0.3, 0.35$ crossed with $b = 1.0, 1.5, 2.0, 2.5, 3.0$, and on a narrow grid, $a = 0.2, 0.25, 0.3$ crossed with $b = 1.5, 2, 2.5$; recording (Philharmonia vs Iowa clarinet); extraction constants (prominence 0.005–0.0002, cap 15 or 25); audio codec (lossless to 48 kbit/s MP3); idealised-spectrum constants (5, 7 or 11 harmonics, roll-off 0.8–1.0). Under the narrow grid the sign of the H&K difference is stable for 11 of 17 spectra. Vassilakis's near-flat amplitude weight makes it depend on how many weak partials are admitted (extraction moves it by up to 168.3 pp; Table S21).

Clarinet. Table S19 tests the H&K clarinet reversal against every constant. It survives the extraction sweep (range at most 1.5 pp), the cutoff sweep, ERB substitution and the whole narrow a, b grid on both libraries (Table S20). At the minor sixth the fifth harmonic of the lower note (1304 Hz) and the third of the upper (1242 Hz) lie near the peak of the H&K kernel; at the tritone the fundamentals are 1.49 critical bandwidths apart, where the H&K weight is 0.17% of its peak but the Sethares kernel still assigns substantial roughness (36% of its cross-tone total). Listener ratings of clarinet dyads at several registers would discriminate between kernels.

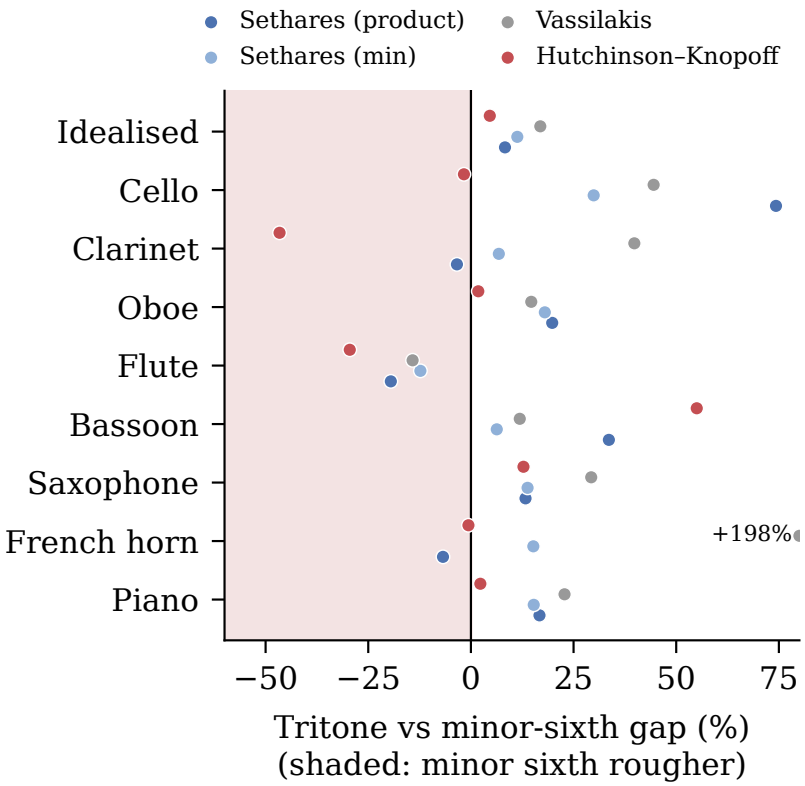


Figure S4: Tritone/minor-sixth roughness difference (%) per spectrum and kernel at C4 (shaded: the kernel rates the minor sixth rougher).

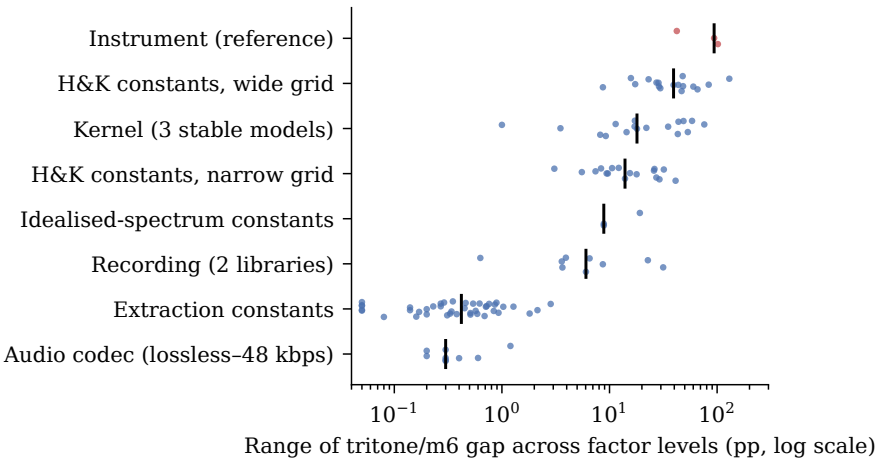


Figure S5: Range of the tritone/minor-sixth difference (pp, log scale) across the levels of each analysis factor, stable kernels; bars, medians. Red: instrument (reference).

Table S13: Peak and trough locations (cents) [participant-bootstrap CI; pooled rows, random effects with Hartung–Knapp CI], kernel SD 0.1 semitones (exploratory). Equal-tempered and just-intoned positions: m6 800/814; TT 600/583 (7/5) or 590 (45/32); P5 700/702; M3 400/386; M6 900/884. †Point estimate on the edge of the search window. Edge: extremum on the window edge in at least one spectrum.

	m6 peak	TT trough	P5 peak	M3 peak	M6 peak
Listeners					
Harmonic, 3 dB/oct	806 [777, 825]	579 [550, 650]	700 [691, 708]	404 [378, 413]	894 [886, 900]
5 equal harmonics	815 [808, 824]	633 [550, 650]	696 [689, 715]	394 [388, 399]	880 [876, 897]
5 harmonics, no 3rd	850 [788, 850]†	552 [550, 650]	693 [686, 708]	387 [380, 398]	899 [857, 907]
Flute	801 [773, 813]	603 [554, 650]	702 [694, 708]	417 [379, 423]	898 [861, 906]
Guitar	810 [781, 818]	650 [562, 650]†	708 [699, 719]	392 [382, 401]	889 [881, 894]
Piano	799 [789, 809]	644 [634, 648]	700 [693, 706]	406 [382, 411]	888 [880, 898]
Pure tones	806 [750, 850]	633 [565, 641]	705 [691, 737]	401 [360, 450]	887 [866, 894]
Stretched (2.1)	750 [750, 755]†	650 [560, 650]†	744 [729, 750]	408 [355, 449]	941 [856, 950]
Compressed (1.9)	824 [750, 850]	550 [550, 627]†	650 [650, 750]†	377 [359, 416]	850 [850, 938]†
Pooled, 6 harmonic (RE)	808 [796, 819]	616 [576, 655]	700 [696, 705]	396 [387, 404]	890 [883, 896]
H&K + harmonicity (6.83 cents)					
Harmonic, 3 dB/oct	814 [814, 815]	644 [643, 645]	703 [702, 704]	387 [387, 388]	886 [885, 887]
5 equal harmonics	814 [813, 815]	588 [586, 589]	701 [701, 702]	385 [384, 386]	885 [884, 886]
5 harmonics, no 3rd	812 [812, 813]	550 [550, 550]†	701 [700, 701]	387 [387, 388]	881 [881, 882]
Flute	815 [814, 815]	632 [631, 633]	703 [702, 704]	388 [388, 389]	887 [886, 887]
Guitar	812 [812, 813]	650 [649, 650]†	703 [702, 704]	390 [389, 390]	882 [881, 883]
Piano	812 [812, 813]	649 [648, 650]	704 [703, 704]	389 [388, 390]	886 [885, 887]
Pure tones	813 [812, 814]	600 [600, 601]	702 [701, 702]	390 [389, 391]	950 [950, 950]†
Stretched (2.1)	751 [750, 752]	650 [650, 650]†	750 [750, 750]†	407 [406, 409]	949 [948, 950]
Compressed (1.9)	813 [812, 814]	595 [594, 596]	650 [650, 650]†	450 [390, 450]†	890 [889, 891]
Pooled, 6 harmonic (RE)	813 [812, 815]	edge	702 [701, 704]	388 [386, 389]	885 [882, 887]
H&K + harmonicity (20 cents)					
Harmonic, 3 dB/oct	850 [850, 850]†	624 [621, 627]	703 [703, 704]	387 [386, 388]	885 [884, 886]
5 equal harmonics	850 [850, 850]†	593 [592, 594]	702 [702, 703]	381 [381, 382]	883 [882, 883]
5 harmonics, no 3rd	810 [809, 810]	550 [550, 550]†	750 [750, 750]†	387 [387, 388]	850 [850, 850]†
Flute	750 [750, 750]†	610 [609, 612]	703 [702, 704]	450 [450, 450]†	888 [887, 888]
Guitar	750 [750, 750]†	601 [601, 602]	706 [705, 707]	450 [450, 450]†	850 [850, 850]†
Piano	750 [750, 750]†	605 [604, 606]	707 [706, 707]	450 [450, 450]†	850 [850, 850]†
Pure tones	750 [750, 750]†	600 [600, 601]	701 [700, 701]	450 [450, 450]†	950 [950, 950]†
Stretched (2.1)	753 [752, 753]	650 [650, 650]†	750 [750, 750]†	411 [411, 412]	947 [946, 948]
Compressed (1.9)	818 [817, 818]	586 [585, 587]	651 [650, 651]	450 [450, 450]†	905 [850, 906]
Pooled, 6 harmonic (RE)	edge	edge	edge	edge	edge
Composite					
Harmonic, 3 dB/oct	814 [813, 814]	649 [648, 650]	703 [702, 704]	387 [386, 388]	886 [885, 887]
5 equal harmonics	813 [813, 814]	592 [591, 592]	701 [701, 702]	386 [385, 387]	885 [884, 885]
5 harmonics, no 3rd	812 [812, 813]	550 [550, 550]†	701 [700, 701]	387 [387, 388]	882 [882, 883]
Flute	814 [813, 815]	635 [634, 635]	703 [702, 704]	388 [387, 389]	886 [885, 887]
Guitar	813 [812, 813]	648 [648, 649]	703 [702, 704]	388 [388, 389]	883 [882, 950]
Piano	813 [812, 813]	649 [648, 649]	704 [703, 705]	388 [387, 389]	886 [885, 887]
Pure tones	813 [812, 814]	600 [600, 601]	702 [701, 702]	389 [388, 390]	950 [950, 950]†
Stretched (2.1)	751 [750, 753]	650 [649, 650]†	750 [750, 750]†	410 [408, 412]	948 [947, 949]
Compressed (1.9)	813 [812, 814]	599 [598, 600]	650 [650, 650]†	450 [450, 450]†	942 [892, 944]
Pooled, 6 harmonic (RE)	813 [812, 814]	edge	702 [701, 704]	387 [386, 388]	885 [883, 887]

Table S14: Gap at equal-tempered (ET) and just-intoned positions (JIa: 8/5 vs 7/5; JIb: 8/5 vs 45/32), observed and left by each held-out model, random effects over the six harmonic spectra [Hartung–Knapp CI], at three kernel widths (semitones; exploratory). A kernel of 0.2 semitones cannot separate positions 14 cents apart.

SD		ET	JIa	JIb	JIa − ET	JIb − ET
0.1	Observed	+0.28 [+0.13, +0.43]	+0.29 [+0.18, +0.40]	+0.27 [+0.17, +0.36]	+0.03 [−0.12, +0.17]	+0.01 [−0.11, +0.12]
	Left by H&K	+0.28 [+0.13, +0.43]	+0.29 [+0.17, +0.41]	+0.27 [+0.16, +0.37]	+0.02 [−0.12, +0.17]	0.00 [−0.11, +0.12]
	Left by H&K + H (6.83)	+0.20 [+0.02, +0.38]	+0.18 [+0.02, +0.34]	+0.16 [+0.02, +0.30]	−0.01 [−0.15, +0.13]	−0.03 [−0.14, +0.08]
	Left by H&K + H (20)	+0.20 [+0.02, +0.38]	+0.22 [+0.06, +0.37]	+0.19 [+0.05, +0.33]	+0.03 [−0.11, +0.17]	+0.01 [−0.10, +0.12]
	Left by composite	+0.17 [−0.03, +0.37]	+0.13 [−0.05, +0.31]	+0.11 [−0.05, +0.28]	−0.03 [−0.17, +0.12]	−0.04 [−0.15, +0.07]
0.05	Observed	+0.32 [+0.08, +0.56]	+0.35 [+0.20, +0.50]	+0.26 [+0.16, +0.36]	+0.05 [−0.25, +0.34]	−0.04 [−0.29, +0.21]
	Left by H&K	+0.32 [+0.08, +0.57]	+0.34 [+0.19, +0.50]	+0.25 [+0.14, +0.36]	+0.04 [−0.25, +0.33]	−0.05 [−0.29, +0.20]
	Left by H&K + H (6.83)	+0.24 [−0.04, +0.51]	+0.21 [+0.02, +0.39]	+0.11 [−0.03, +0.25]	0.00 [−0.29, +0.28]	−0.09 [−0.33, +0.15]
	Left by H&K + H (20)	+0.24 [−0.04, +0.51]	+0.26 [+0.09, +0.44]	+0.16 [+0.02, +0.29]	+0.05 [−0.23, +0.34]	−0.04 [−0.28, +0.20]
	Left by composite	+0.20 [−0.10, +0.50]	+0.15 [−0.05, +0.35]	+0.06 [−0.11, +0.22]	−0.03 [−0.31, +0.26]	−0.11 [−0.34, +0.13]
0.2	Observed	+0.26 [+0.15, +0.36]	+0.25 [+0.16, +0.34]	+0.25 [+0.16, +0.33]	0.00 [−0.05, +0.04]	−0.01 [−0.04, +0.03]
	Left by H&K	+0.26 [+0.14, +0.37]	+0.25 [+0.15, +0.35]	+0.24 [+0.15, +0.34]	−0.01 [−0.05, +0.04]	−0.01 [−0.05, +0.02]
	Left by H&K + H (6.83)	+0.19 [+0.06, +0.33]	+0.18 [+0.05, +0.30]	+0.17 [+0.05, +0.29]	−0.02 [−0.06, +0.02]	−0.02 [−0.05, +0.01]
	Left by H&K + H (20)	+0.18 [+0.04, +0.32]	+0.18 [+0.05, +0.32]	+0.17 [+0.04, +0.30]	0.00 [−0.04, +0.04]	−0.01 [−0.04, +0.02]
	Left by composite	+0.17 [+0.02, +0.32]	+0.15 [+0.01, +0.29]	+0.14 [+0.01, +0.28]	−0.02 [−0.06, +0.02]	−0.02 [−0.05, +0.01]

Table S15: Unique ΔR^2 [95% bootstrap CI] (p, nested F) of interference (H&K), harmonicity and familiarity over the other two, under four covariate specifications. R^2: full model including covariates.

Covariates	R^2	Interference	Harmonicity	Familiarity
Bowling et al. (2018), $n = 298$				
None (as originally reported)	0.674	0.049 [0.026, 0.080] (<.001)	0.010 [0.001, 0.028] (.004)	0.147 [0.096, 0.204] (<.001)
Chord size	0.738	0.075 [0.045, 0.109] (<.001)	0.037 [0.017, 0.065] (<.001)	0.080 [0.043, 0.122] (<.001)
Chord size + lowest note	0.749	0.085 [0.053, 0.122] (<.001)	0.042 [0.019, 0.068] (<.001)	0.058 [0.030, 0.091] (<.001)
Chord size; models on transposed chords	0.737	0.074 [0.046, 0.109] (<.001)	0.043 [0.020, 0.072] (<.001)	0.082 [0.045, 0.124] (<.001)
Bowling et al. (2018), interference and harmonicity rescored at the presented just-intonation pitches				
None (as originally reported)	0.692	0.038 [0.017, 0.065] (<.001)	0.024 [0.008, 0.050] (<.001)	0.133 [0.086, 0.184] (<.001)
Chord size	0.775	0.069 [0.040, 0.102] (<.001)	0.068 [0.041, 0.101] (<.001)	0.061 [0.032, 0.095] (<.001)
Chord size + lowest note	0.786	0.080 [0.048, 0.117] (<.001)	0.074 [0.045, 0.104] (<.001)	0.043 [0.021, 0.068] (<.001)
Chord size; models on transposed chords	0.775	0.069 [0.042, 0.104] (<.001)	0.071 [0.042, 0.104] (<.001)	0.064 [0.035, 0.097] (<.001)
Johnson-Laird et al. (2012), $n = 103$				
None (as originally reported)	0.414	0.005 [0.000, 0.051] (.344)	0.042 [0.009, 0.101] (.009)	0.141 [0.046, 0.263] (<.001)
Chord size	0.500	0.033 [0.001, 0.095] (.013)	0.100 [0.039, 0.180] (<.001)	0.045 [0.004, 0.117] (.004)
Chord size + lowest note	0.516	0.026 [0.000, 0.090] (.025)	0.108 [0.044, 0.186] (<.001)	0.033 [0.002, 0.090] (.011)
Chord size; models on transposed chords	0.534	0.068 [0.019, 0.143] (<.001)	0.090 [0.041, 0.168] (<.001)	0.031 [0.001, 0.088] (.012)

Table S16: Tritone/minor-sixth roughness difference (%) at C4. Bold: negative (the kernel rates the minor sixth rougher). Vassilakis values are highly sensitive to extraction settings and should be read for sign only.

Spectrum	Sethares	Sethares-min	Vassilakis	H&K
Idealised (7 harmonics, 0.9^{n-1})	+8.3	+11.3	+16.9	+4.6
Cello	+74.3	+29.9	+44.5	**−1.7**
Clarinet	**−3.4**	+6.8	+39.8	**−46.6**
Oboe	+19.8	+18.0	+14.7	+1.8
Flute	**−19.5**	**−12.3**	**−14.2**	**−29.5**
Bassoon	+33.6	+6.3	+11.9	+55.0
Saxophone	+13.3	+13.8	+29.3	+12.8
French horn	**−6.8**	+15.2	+198.1	**−0.6**
Piano (Iowa)	+16.7	+15.3	+22.8	+2.3
Negative gaps (of 8 measured)	3	1	1	4

Table S17: Tritone/minor-sixth roughness difference (%) at three registers (Philharmonia). Bold: negative.

Spectrum	Sethares	Sethares-min	Vassilakis	H&K
Clarinet D3	+12.3	+11.9	+21.8	**−30.8**
Clarinet C4	**−3.4**	+6.8	+39.8	**−46.6**
Clarinet C5	+15.3	+4.0	+65.9	**−43.2**
Cello C3	+7.6	+11.1	+13.8	+9.5
Cello C4	+74.3	+29.9	+44.5	**−1.7**
Cello C5	**−6.9**	+1.3	+46.9	**−5.0**
Oboe A♯3	**−9.1**	**−4.5**	+8.3	**−21.6**
Oboe C4	+19.8	+18.0	+14.7	+1.8
Oboe C5	+18.7	+23.1	+179.7	+11.7

Table S18: Range of the tritone/minor-sixth roughness difference (pp) across the levels of each analysis factor. n: number of (spectrum, kernel) cells for the three stable kernels. For the instrument row each cell is one kernel across the eight C4 instruments. †H&K only. ‡For Vassilakis this row shows its amplitude exponent varied over $\{0.1, 0.25, 0.5, 1\}$.

		Stable kernels		Vassilakis	
Factor	n	Median	Max	Median	Max
Instrument (8 at C4; reference quantity)	3	93.8	101.6	212.3	212.3
Kernel (Sethares, Sethares-min, H&K)	17	18.0	76.0	48.7	204.9
H&K constants a, b, wide grid†	17	39.4	129.8	24.5	473.9‡
H&K constants a, b, narrow grid†	17	13.9	41.1	–	–
Idealised-spectrum constants (partials, roll-off)	3	8.8	19.1	24.6	24.6
Recording (Philharmonia vs Iowa clarinet)	9	6.0	31.5	23.6	51.7
Extraction constants (prominence, cap)	42	0.4	2.8	9.3	168.3
Audio codec (lossless to 48 kbit/s)	9	0.3	1.2	4.3	5.8

Table S19: Robustness of the H&K clarinet difference (%). Extraction: range (pp) across the extraction sweep. Cutoff: range with the 1.2-CBW cutoff at 1.2, 1.5, 2.0 or removed (Philharmonia only). Narrow/wide: range over the a, b grids. ERB: difference with the ERB in place of the CBW. Bold: positive.

Reg.	Library	Diff.	Extraction	Cutoff	Narrow a, b	Wide a, b	ERB
D3	Philharmonia	−30.8	0.54	[−30.8, −29.0]	[−40.6, −13.3]	[−45.1, **+1.6**]	−36.5
D3	Iowa	−22.1	1.46	–	[−32.7, −6.5]	[−35.3, **+4.1**]	−26.2
C4	Philharmonia	−46.6	0.27	[−46.6, −45.6]	[−50.7, −38.5]	[−54.0, −10.4]	−46.5
C4	Iowa	−43.0	0.03	–	[−47.1, −33.2]	[−49.6, −19.9]	−46.6
C5	Philharmonia	−43.2	0.33	[−43.2, −42.5]	[−45.2, −39.7]	[−47.1, −19.8]	−46.3
C5	Iowa	−11.7	0.03	–	[−13.2, −10.2]	[−14.0, −5.3]	−15.7

Table S20: H&K tritone/minor-sixth difference (%) per spectrum under the a, b grids and ERB substitution. Bold: sign differs from the baseline.

Spectrum	Baseline	Narrow a, b	Wide a, b	ERB, cut 1.2	ERB, no cut
Bassoon C4 (Phil.)	+55.0	[+43.3, +58.8]	[+14.0, +61.9]	+54.2	+51.9
Cello C3 (Phil.)	+9.5	[−3.9, +14.0]	[−14.0, +14.8]	+12.1	+12.7
Cello C4 (Phil.)	−1.7	[−30.1, +11.0]	[−55.3, +28.2]	**+7.0**	**+5.8**
Cello C5 (Phil.)	−5.0	[−9.7, −2.3]	[−16.4, −0.6]	−7.9	−8.0
Clarinet C4 (Phil.)	−46.6	[−50.7, −38.5]	[−54.0, −10.4]	−46.5	−45.6
Clarinet C5 (Phil.)	−43.2	[−45.2, −39.7]	[−47.1, −19.8]	−46.3	−46.0
Clarinet D3 (Phil.)	−30.8	[−40.6, −13.3]	[−45.1, +1.6]	−36.5	−35.2
Flute C4 (Phil.)	−29.5	[−33.0, −22.4]	[−35.8, −12.7]	−33.5	−33.1
French horn C4 (Phil.)	−0.6	[−9.4, +0.2]	[−27.5, +1.0]	−0.7	−1.1
Oboe A♯3 (Phil.)	−21.6	[−25.1, −15.7]	[−27.2, +21.1]	−20.8	−20.0
Oboe C4 (Phil.)	+1.8	[−10.3, +21.6]	[−17.1, +112.8]	**−0.2**	+0.3
Oboe C5 (Phil.)	+11.7	[−6.0, +23.2]	[−30.1, +30.0]	+10.2	+10.1
Saxophone C4 (Phil.)	+12.8	[+6.6, +15.0]	[−0.8, +16.5]	+13.0	+12.6
Piano C4 (Iowa)	+2.3	[−16.9, +9.0]	[−39.5, +26.3]	+4.2	+3.4
Clarinet D3 (Iowa)	−22.1	[−32.7, −6.5]	[−35.3, +4.1]	−26.2	−25.3
Clarinet C4 (Iowa)	−43.0	[−47.1, −33.2]	[−49.6, −19.9]	−46.6	−45.8
Clarinet C5 (Iowa)	−11.7	[−13.2, −10.2]	[−14.0, −5.3]	−15.7	−15.7

Table S21: Vassilakis tritone/minor-sixth difference (%) as the amplitude exponent varies (published value 0.1). Bold: sign differs from the published-exponent value.

Spectrum	0.1	0.25	0.5	1.0
Bassoon C4 (Phil.)	+11.9	+4.6	**−3.9**	+8.6
Cello C3 (Phil.)	+13.8	+16.5	+18.6	+16.1
Cello C4 (Phil.)	+44.5	+49.5	+76.0	+228.8
Cello C5 (Phil.)	+46.9	+34.2	+10.2	**−4.0**
Clarinet C4 (Phil.)	+39.8	+38.5	+37.5	+65.8
Clarinet C5 (Phil.)	+65.9	+69.9	+94.5	+539.9
Clarinet D3 (Phil.)	+21.8	+24.6	+32.7	+59.7
Flute C4 (Phil.)	−14.2	−18.9	−26.2	−37.5
French horn C4 (Phil.)	+198.1	+182.2	+175.5	+200.0
Oboe A♯3 (Phil.)	+8.3	**−2.0**	**−14.8**	**−23.4**
Oboe C4 (Phil.)	+14.7	+14.7	+17.0	+36.7
Oboe C5 (Phil.)	+179.7	+102.2	+32.0	+3.1
Saxophone C4 (Phil.)	+29.3	+28.4	+28.3	+36.8
Piano C4 (Iowa)	+22.8	+23.6	+31.3	+84.2
Clarinet D3 (Iowa)	+3.4	+2.9	+3.8	+14.3
Clarinet C4 (Iowa)	+16.2	+8.9	+3.7	+27.6
Clarinet C5 (Iowa)	+14.2	+9.7	+6.1	+12.9

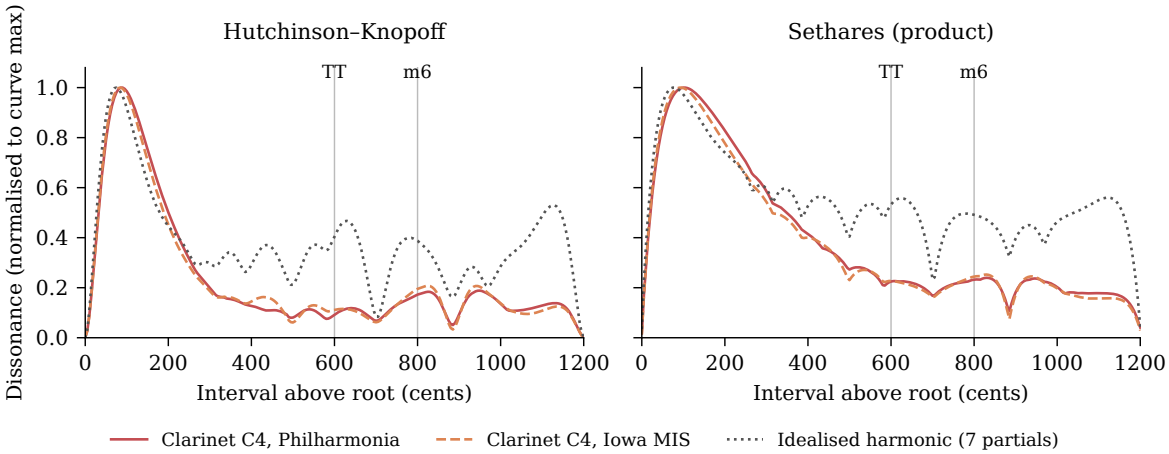


Figure S6: Dissonance curves (normalised to each curve's maximum) for two clarinet C4 recordings and an idealised seven-partial harmonic tone, under the H&K and Sethares (product) kernels.

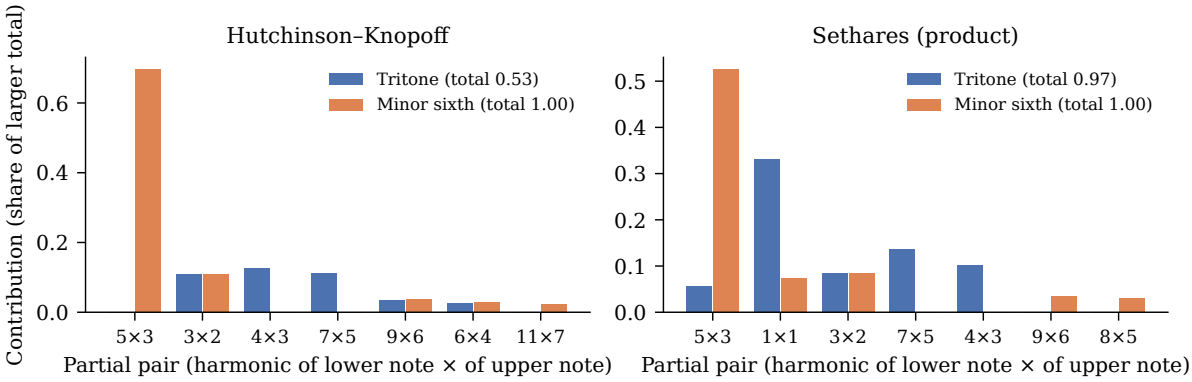


Figure S7: Philharmonia clarinet C4: the partial pairs contributing most roughness at the tritone and the minor sixth under each kernel (share of the larger total).

S16 The main analyses, experiment by experiment

The main text reports results combined over experiments. The tables of this section give the same analyses for every experiment and every model; all are described in the Method and Results sections of the main text. Table S22 gives the gap in each experiment, observed and left unexplained by each curve fitted to the experiment itself. Table S23 gives the pooled values of those fits. Table S24 gives the gap and dip left when each experiment is predicted with weights fitted to the others. Table S25 gives the held-out residual at every whole-number interval. Table S26 gives the differences between groups of listeners and between spectra behind the four targeted tests. Table S27 gives the exploratory decomposition of the held-out gap into an in-tune bonus and a step (main-text Section 8.1).

Table S22: The gap (minor sixth minus tritone), observed and left unexplained by each model fitted to the experiment itself, in within-listener SD [95% participant-bootstrap CI]. Bold: CI excludes zero. The composite of Marjieh et al. (2024) was fitted to these data. N: participants analysed.

			Left unexplained after		
Experiment	N	Observed	Roughness	Roughness + harmonicity	Composite
United States					
Harmonic, 3 dB/oct	160	**+0.19 [+0.04, +0.35]**	**+0.18 [+0.03, +0.33]**	+0.12 [−0.03, +0.28]	**+0.16 [+0.01, +0.31]**
5 equal harmonics	147	**+0.18 [+0.07, +0.30]**	**+0.19 [+0.07, +0.31]**	+0.04 [−0.08, +0.16]	+0.05 [−0.06, +0.16]
5 harmonics, no 3rd	151	**+0.16 [+0.06, +0.26]**	**+0.15 [+0.05, +0.25]**	+0.01 [−0.10, +0.12]	0.00 [−0.10, +0.10]
Flute	188	**+0.35 [+0.22, +0.48]**	**+0.36 [+0.23, +0.50]**	**+0.34 [+0.20, +0.47]**	**+0.27 [+0.13, +0.40]**
Guitar	186	**+0.39 [+0.25, +0.55]**	**+0.40 [+0.25, +0.55]**	**+0.37 [+0.23, +0.52]**	**+0.33 [+0.19, +0.48]**
Piano	191	**+0.33 [+0.18, +0.48]**	**+0.32 [+0.18, +0.48]**	**+0.29 [+0.15, +0.45]**	**+0.25 [+0.11, +0.40]**
Pure tones	151	**+0.21 [+0.08, +0.34]**	**+0.21 [+0.08, +0.34]**	**+0.18 [+0.05, +0.31]**	**+0.14 [+0.02, +0.26]**
Bonang (harmonic bass)	146	**+0.38 [+0.22, +0.53]**	**+0.29 [+0.14, +0.44]**	**+0.31 [+0.16, +0.46]**	**+0.28 [+0.12, +0.42]**
Stretched (octave 2.1)	153	−0.02 [−0.17, +0.11]	−0.02 [−0.17, +0.11]	−0.02 [−0.17, +0.11]	−0.02 [−0.16, +0.12]
Compressed (octave 1.9)	154	+0.02 [−0.12, +0.14]	−0.02 [−0.15, +0.10]	−0.02 [−0.15, +0.11]	−0.04 [−0.17, +0.08]
South Korea					
Harmonic, 3 dB/oct	40	+0.07 [−0.12, +0.25]	+0.06 [−0.12, +0.24]	−0.03 [−0.20, +0.14]	+0.03 [−0.14, +0.21]
Stretched (octave 2.1)	37	−0.17 [−0.34, 0.00]	−0.17 [−0.34, 0.00]	**−0.18 [−0.34, 0.00]**	−0.17 [−0.34, 0.00]
Compressed (octave 1.9)	45	**−0.21 [−0.39, −0.04]**	**−0.18 [−0.35, −0.02]**	−0.14 [−0.31, +0.03]	**−0.16 [−0.33, 0.00]**

Table S23: Random-effects pooled estimates [95% Hartung–Knapp CI] and 95% prediction intervals (PI) for a new experiment, SD units, with each model fitted to the experiment itself. Harmonic: the six US experiments with harmonic spectra. Eight: adds pure tones and bonang. [a]Fitted to these data. DerSimonian–Laird intervals: Table S4.

	Gap, harmonic ($k = 6$)		Gap, eight ($k = 8$)		Dip, harmonic ($k = 6$)	
	Estimate [CI]	PI	Estimate [CI]	PI	Estimate [CI]	PI
Observed	+0.26 [+0.16, +0.36]	[+0.03, +0.49]	+0.27 [+0.18, +0.35]	[+0.08, +0.45]	+0.30 [+0.22, +0.37]	[+0.17, +0.43]
After roughness (H&K)	+0.26 [+0.15, +0.37]	[−0.01, +0.53]	+0.26 [+0.18, +0.33]	[+0.07, +0.44]	+0.25 [+0.13, +0.36]	[−0.03, +0.52]
After roughness (Sethares)	+0.25 [+0.13, +0.38]	[−0.05, +0.56]	+0.25 [+0.16, +0.34]	[+0.01, +0.49]	+0.28 [+0.17, +0.39]	[+0.03, +0.53]
After harmonicity alone	+0.11 [−0.05, +0.27]	[−0.32, +0.55]	+0.14 [+0.02, +0.27]	[−0.22, +0.51]	+0.03 [−0.10, +0.16]	[−0.31, +0.37]
After roughness + harmonicity	+0.19 [+0.02, +0.36]	[−0.26, +0.64]	+0.20 [+0.08, +0.32]	[−0.14, +0.55]	+0.14 [−0.02, +0.29]	[−0.25, +0.53]
harmonicity template $i^{-0.75}$ (`incon`)	+0.20 [+0.05, +0.36]	[−0.21, +0.61]	+0.21 [+0.10, +0.32]	[−0.10, +0.52]	+0.15 [0.00, +0.30]	[−0.23, +0.53]
After revised H&K + harmonicity	+0.28 [+0.11, +0.44]	[−0.16, +0.71]	+0.27 [+0.15, +0.38]	[−0.05, +0.58]	+0.20 [+0.02, +0.39]	[−0.28, +0.68]
After Marjieh et al. composite[a]	+0.17 [+0.03, +0.31]	[−0.20, +0.54]	+0.18 [+0.08, +0.28]	[−0.11, +0.46]	+0.05 [−0.13, +0.23]	[−0.43, +0.54]

Table S24: Leave-one-experiment-out prediction (published tolerance). b_R, b_H: roughness and harmonicity weights fitted to the other US experiments. Held-out unexplained gap and dip [95% CI], SD units. Bold: CI excludes zero.

Held-out experiment	b_R	b_H	Gap, roughness only	Gap, two mechanisms	Dip, two mechanisms
Harmonic, 3 dB/oct	−1.66	0.63	**+0.18 [+0.03, +0.33]**	+0.15 [0.00, +0.30]	**+0.15 [+0.02, +0.29]**
5 equal harmonics	−1.62	0.63	**+0.19 [+0.07, +0.30]**	+0.09 [−0.02, +0.21]	−0.02 [−0.15, +0.11]
5 harmonics, no 3rd	−1.61	0.64	**+0.15 [+0.05, +0.25]**	+0.04 [−0.06, +0.14]	**+0.24 [+0.15, +0.34]**
Flute	−1.52	0.70	**+0.36 [+0.23, +0.49]**	**+0.30 [+0.17, +0.44]**	−0.01 [−0.13, +0.11]
Guitar	−1.54	0.70	**+0.40 [+0.25, +0.55]**	**+0.35 [+0.21, +0.50]**	**+0.21 [+0.08, +0.35]**
Piano	−1.52	0.69	**+0.32 [+0.18, +0.48]**	**+0.28 [+0.14, +0.44]**	**+0.25 [+0.14, +0.38]**
Pure tones	−1.55	0.71	**+0.21 [+0.08, +0.34]**	**+0.17 [+0.04, +0.30]**	**+0.17 [+0.06, +0.29]**
Bonang (harmonic bass)	−1.68	0.66	**+0.21 [+0.05, +0.36]**	**+0.19 [+0.03, +0.34]**	+0.09 [−0.04, +0.22]
Stretched (octave 2.1)	−1.63	0.66	−0.03 [−0.17, +0.11]	−0.02 [−0.16, +0.11]	−0.04 [−0.15, +0.08]
Compressed (octave 1.9)	−1.66	0.65	−0.06 [−0.20, +0.06]	−0.06 [−0.19, +0.06]	+0.01 [−0.10, +0.12]

Table S25: Held-out residual (observed − predicted) at each integer interval, random effects over the six harmonic US spectra, SD units. *Hartung–Knapp CI excludes zero. Size: a linear term in interval size (main-text Section 7.8). Columns: m minor, M major, P perfect; 2 second, 3 third, 4 fourth, 5 fifth, 6 sixth, 7 seventh, 8 octave, 9 ninth; TT tritone.

Model	m2	M2	m3	M3	P4	TT	P5	m6	M6	m7	M7	P8	m9	M9
H&K	−0.01	−0.10	+0.09	+0.19*	+0.14*	−0.09	+0.16*	+0.17*	+0.15*	0.00	+0.03	+0.25*	−0.12	−0.08
H&K + H (6.83)	−0.02	−0.14	+0.11	+0.18*	+0.07	−0.05	+0.10*	+0.16*	+0.18*	−0.02	+0.07	0.00	−0.08	−0.09
H&K + H (20)	+0.01	−0.09	+0.12*	+0.19*	+0.11	−0.02	+0.14*	+0.17*	+0.19*	+0.03	+0.09*	+0.04	−0.07	−0.05
H&K + H (20) + size	−0.01	−0.16*	+0.04	+0.11*	+0.04	−0.06	+0.09*	+0.15*	+0.18*	+0.05	+0.16*	+0.07	+0.04	+0.03

Table S26: Differences between groups [95% bootstrap CI], SD units. Musicianship rows give the number of musicians/non-musicians. Bold: CI excludes zero (uncorrected).

	Observed gap	Gap after rough. + harm.	Observed dip	Dip after rough. + harm.
Korean minus US listeners, same stimuli				
Harmonic	−0.13 [−0.37, +0.10]	−0.14 [−0.39, +0.08]	−0.13 [−0.31, +0.07]	−0.14 [−0.32, +0.03]
Stretched	−0.15 [−0.36, +0.07]	−0.16 [−0.36, +0.07]	−0.03 [−0.22, +0.17]	−0.02 [−0.21, +0.18]
Compressed	**−0.23 [−0.45, −0.02]**	−0.12 [−0.34, +0.09]	−0.04 [−0.22, +0.17]	+0.01 [−0.19, +0.20]
Musicians minus non-musicians (US)				
Harmonic, 3 dB/oct (69/91)	+0.17 [−0.13, +0.44]	+0.13 [−0.19, +0.41]	**+0.36 [+0.10, +0.61]**	+0.18 [−0.05, +0.40]
5 equal harmonics (63/84)	+0.08 [−0.15, +0.33]	−0.10 [−0.36, +0.17]	**+0.26 [+0.01, +0.53]**	−0.05 [−0.28, +0.22]
5 harmonics, no 3rd (64/87)	+0.10 [−0.11, +0.30]	+0.01 [−0.20, +0.21]	**+0.29 [+0.10, +0.47]**	**+0.27 [+0.07, +0.45]**
Flute (86/102)	−0.03 [−0.31, +0.23]	−0.08 [−0.36, +0.19]	+0.06 [−0.16, +0.30]	−0.15 [−0.38, +0.10]
Guitar (86/100)	**+0.30 [+0.01, +0.61]**	**+0.28 [0.00, +0.58]**	+0.25 [−0.02, +0.52]	+0.18 [−0.09, +0.44]
Piano (73/117)	**+0.38 [+0.07, +0.66]**	**+0.35 [+0.05, +0.64]**	**+0.37 [+0.12, +0.60]**	**+0.27 [+0.02, +0.50]**
Pure tones (64/87)	+0.18 [−0.07, +0.44]	+0.15 [−0.10, +0.41]	+0.09 [−0.16, +0.32]	+0.01 [−0.24, +0.25]
Bonang (harmonic bass) (71/75)	+0.29 [−0.03, +0.61]	+0.17 [−0.15, +0.48]	**+0.37 [+0.13, +0.62]**	+0.19 [−0.04, +0.42]
Stretched (octave 2.1) (62/91)	0.00 [−0.26, +0.27]	+0.01 [−0.26, +0.28]	+0.04 [−0.20, +0.28]	+0.03 [−0.20, +0.27]
Compressed (octave 1.9) (68/86)	−0.21 [−0.47, +0.07]	−0.20 [−0.46, +0.08]	+0.06 [−0.17, +0.29]	+0.08 [−0.15, +0.31]

Table S27: Decomposition of the held-out gap (exploratory; H&K + harmonicity at 20 cents + interval size), per harmonic spectrum [participant-bootstrap CI] and pooled by random effects [Hartung–Knapp CI]. Bonus part $b(8) - b(6)$; baseline part $q(8) - q(6)$; step: baseline rise between 7 and 8 semitones after a linear trend; other − TT: mean bonus of the other intervals minus the tritone's. Bold: CI excludes zero. Guitar and piano use proxy spectra.

Spectrum	Gap left	Bonus part	Baseline part	Step	Other − TT bonus
Harmonic, 3 dB/oct	+0.14 [0.00, +0.29]	+0.13 [−0.02, +0.27]	+0.01 [−0.08, +0.11]	−0.02 [−0.12, +0.07]	+0.09 [−0.02, +0.19]
5 equal harmonics	+0.10 [−0.01, +0.22]	−0.02 [−0.15, +0.11]	**+0.12 [+0.04, +0.20]**	+0.04 [−0.03, +0.10]	−0.05 [−0.15, +0.04]
5 harmonics, no 3rd	+0.05 [−0.05, +0.15]	−0.02 [−0.12, +0.10]	+0.07 [−0.02, +0.16]	+0.09 [0.00, +0.17]	**+0.12 [+0.04, +0.20]**
Flute	**+0.33 [+0.19, +0.47]**	**+0.16 [+0.02, +0.29]**	**+0.17 [+0.08, +0.26]**	+0.01 [−0.08, +0.10]	+0.06 [−0.05, +0.16]
Guitar	**+0.39 [+0.23, +0.55]**	0.00 [−0.16, +0.17]	**+0.39 [+0.28, +0.48]**	**+0.31 [+0.22, +0.41]**	+0.04 [−0.08, +0.15]
Piano	**+0.30 [+0.15, +0.45]**	**+0.16 [+0.03, +0.29]**	**+0.14 [+0.04, +0.24]**	**+0.17 [+0.08, +0.27]**	**+0.15 [+0.05, +0.25]**
Pooled (RE, Hartung–Knapp)	**+0.21 [+0.06, +0.36]**	+0.06 [−0.03, +0.16]	**+0.15 [+0.02, +0.28]**	+0.10 [−0.03, +0.23]	+0.07 [−0.01, +0.14]
I^2 (%)	77	43	85	87	53

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